

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, CAD/CAM)

December 2012

ME – 701.01 Computer Aided Design

Date: 22.12.2012, Saturday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator & PSG data book is allowed.

SECTION- I

- Q1 (a)** How does the CAD help to reduce product development cycle? Explain. 5
- (b)** Answer following with correct option. 6
1. Best quality graphics is produced by
(A) Line printers (C) Plotter
(B) Daisy wheel printers (D) Dot-matrix printers
 2. The representation of complex 3-D geometrics of the outer shell of a product is called
(A) 2-D modelling (C) 3-D modelling
(B) Solid modelling (D) Surface modelling
 3. The method of solid modelling that define the topology of faces, edges & vertices is called
(A) Layering (C) CSG
(B) B-rep (D) Isometric
 4. One of the major benefits of using CAD in tool design is that for
(A) Product design (C) Simplified trial and error method
(B) Interface checking (D) Inventory control
 5. Tablet cursors is
(A) Digitizer (C) Light pen
(B) Joystick (D) Programmable dials
 6. A convenient method of representing geometrical transformations is by introducing the
(A) Cartesian coordinate system (C) Polar coordinate system
(B) Homogeneous coordinate system (D) All of the above
- Q2 (a)** Compare the capabilities, advantages and disadvantages of types of displays. 4
- (b)** What is scan conversion? Explain scan conversion of a line. 3
- (c)** A line joining point P (12, 12) and Q (20, 18) is to be drawn using Bresenham's algorithm. Find the coordinates of pixels starting from point P. 5

OR

- Q2 (a)** Differentiate between Analytic curves and Parametric curves. 3
- (b)** Explain: "Bresenham's algorithm is more efficient than the DDA line generation algorithm". 3
- (c)** Write Bresenham's midpoint circle algorithm. Scan convert a circle whose centre is (10, 20) and radius is 10 units. 6

- Q3 (a)** Compare geometric transformation with coordinate transformation. Also explain composite transformation. 3
- (b)** What is homogenous coordinate system? Why is it preferred in computer graphics? 3
- (c)** Consider a triangle ABC having coordinates A (5, 5), B (8, 5) and C (5, 10). Determine the new vertex positions if it is mirrored about a line $y = 2x + 4$. 6

OR

- Q3 (a)** Write the matrices for the following 2-D transformations: 3
(i) Scaling (ii) Rotation (iii) Shearing.
- (b)** Describe the capabilities of a typical 3D – CAD software. 3
- (c)** Consider a triangle ABC having coordinates A (5,5), B (8,5) and C (5,10). Determine the new vertex positions if the triangle is rotated by 60° CCW about the vertex A. 6

SECTION- II

- Q4 (a)** Write down the steps for generation of solid model of a Hexagonal Nut using any solid modeling software. 4
- (b)** Explain parametric continuity conditions for curves. 4
- (c)** Explain feature based models. How are those constructed? 3
- Q5 (a)** Explain rotary mode and plane mode of scanning techniques & list out different reverse engineering software's. 6
- (b)** A Cubic Bezier Curve is defined by points (1, 1) (2, 3) (4, 4) & (6, 1). Calculate the coordinate of parametric midpoint of the curve & verify that its gradient (dy/dx) is $1/7^{\text{th}}$ at this point. 6

OR

- Q5 (a)** What is reverse engineering? Justify its use. 4
- (b)** By giving suitable examples explain the various techniques used for surface modeling. 4
- (c)** Calculate the parametric midpoint of Hermite cubic curve that fits the points $P_0 = (1, 3)$, $P_1 = (7, 2)$ and the tangent vectors $P'_0 = (10, 8)$, $P'_1 = (6, 0)$. 4
- Q6 (a)** Write a short notes on the following: 6
(i) GKS (ii) NURBS
- (b)** Write an interactive program and algorithm for four bar chain mechanism. 6

OR

- Q6 (a)** Discuss the need for standardization in computer graphics & list the various CAD standards. 6
- (b)** Prepare an exhaustive algorithm for design of connecting rod of an IC engine. 6